

HESHAN MADHUWANTHA

Physical Science & ICT Undergraduate | Software & Web Developer

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PROFILE

Physical Science & ICT undergraduate with a strong multidisciplinary foundation in mathematics, electronics, and computer science. Experienced in building full-stack and GUI desktop applications using Java, Python, and web technologies, with hands-on exposure to database design, numerical methods, and software testing. A disciplined, detail-oriented problem solver who brings the same patience and creativity from music practice into writing clean, well-structured code.

EDUCATION

BSc in Physical Science & ICT 2024 — Present
Undergraduate degree covering Mathematics, Physics, Electronics, and Computer Science.

G.C.E. Advanced Level Examination 2020 — 2022
Streams: Mathematics, Physics, Information & Communication Technology.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, C (Basics), R, MATLAB (Basic)

Web Development: HTML, CSS, JavaScript, React (Learning), Java Swing (GUI)

Database & Tools: SQLite, MySQL, Git & GitHub, Maven, NetBeans, JUnit 5, VS Code, Linux/Terminal

Mathematics & Science: Linear Algebra, Numerical Methods, Electronics & Physics

PROJECTS

Sales Management System — Java, Swing, SQLite, MVC, JUnit 5, Maven

- Built a full desktop GUI application following the MVC architecture, with inventory tracking, sales reporting, and customer management.
- Achieved complete JUnit 5 test coverage across core modules and used Maven for dependency and build management.

Numerical Methods Solver — Python

- Developed an interactive solver implementing root-finding (Bisection, Newton-Raphson, Secant) and interpolation methods (Lagrange, Newton's Divided Differences, Hermite) with step-by-step output.

Personal Portfolio Website — HTML, CSS, JavaScript

- Designed and built a multi-page responsive portfolio from scratch with a dark theme, scroll animations, skill progress bars, and a working contact form.

Database Concurrency Study Tool — SQL, HTML

- Created a structured study resource and interactive quiz covering ACID properties, locking mechanisms, MVCC, and deadlock resolution.

INTERESTS

Piano Playing • Vocal Music • Quantum Physics • Electronics • Software Development • Mathematics